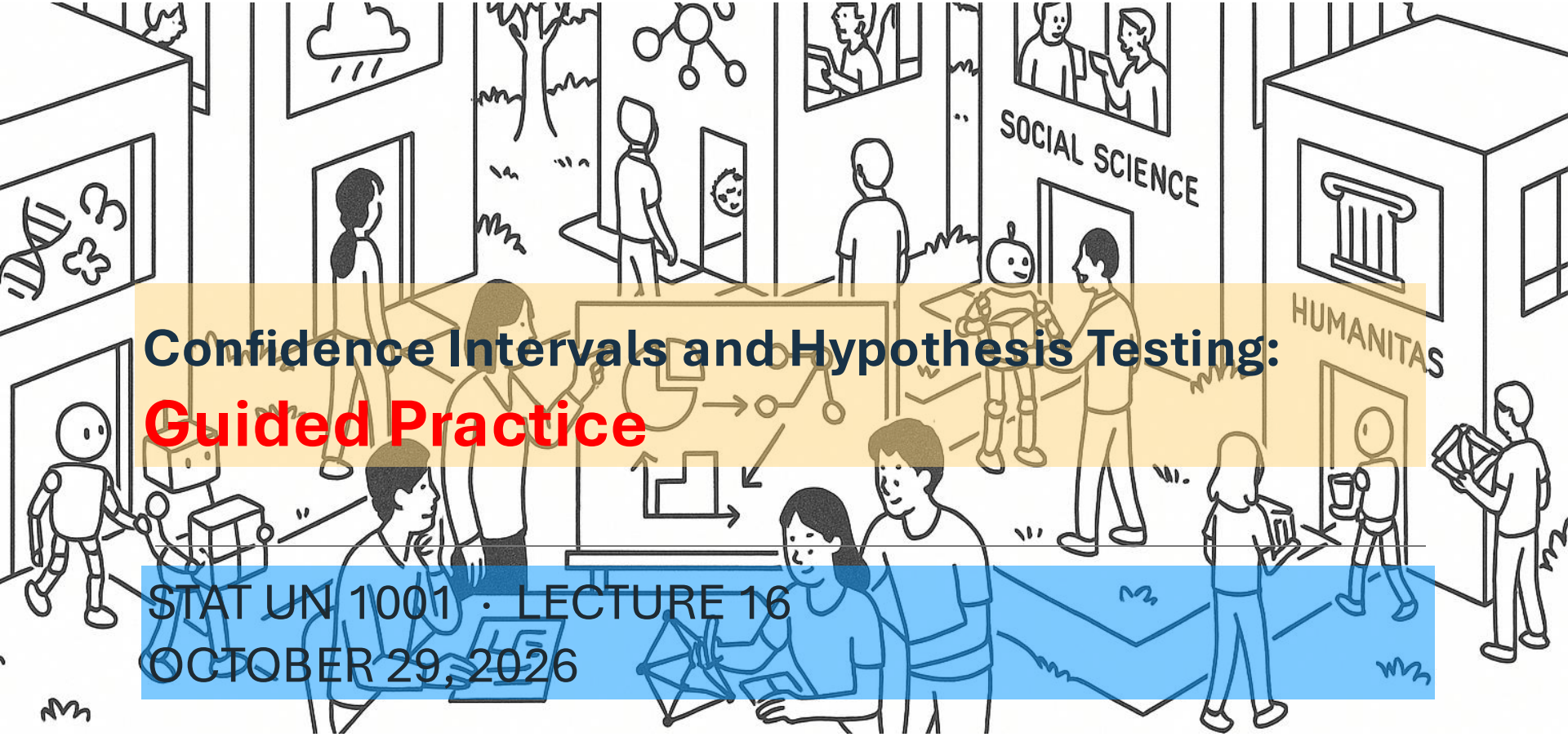


Confidence Intervals and Hypothesis Testing:

Guided Practice

STAT UN 1001 · LECTURE 16

OCTOBER 29, 2026

Practice 1:

Question 1 — 0 pts

A random sample of 1,000 adults gives `p-hat = 0.887`. When the randomization and success-failure conditions are reasonable, what model describes the sampling distribution of `p-hat`?

- ◯ A model for one individual response only
- ◯ Just show me the answer.
- ◯ Uniform, because every proportion is equally likely
- ◯ Approximately normal, centered at the population proportion p
- ◯ Binomial with 1,000 possible values

Practice 1: check the model for $\hat{p}$

A random sample of n = 1,000 adults includes 887 who support a policy. Thus $\hat{p}$ = 887 / 1,000 = 0.887.

Randomness / independence	How was the sample selected? Is it small relative to the population?
Expected successes	$n\hat{p}$ = 1,000(0.887) = 887, which is at least 10.
Expected failures	$n(1-\hat{p})$ = 1,000(0.113) = 113, which is at least 10.

When the conditions fit, we can use an approximate normal model for $\hat{p}$.

Practice 2: Margin of error

Question 6 0 pts

In Example 5.8, `p-hat=0.887`, `SE=0.010`, and `z*=1.96`. What is the margin of error?

- ○ 1.96
- ○ Just show me the answer.
- ○ 0.0196
- ○ 0.887
- ○ 0.010

Margin of error

Example 5.8 provides SE = 0.010 and uses z* = 1.96 for a 95% confidence level.

$$\text{margin of error} = z^* \times SE = 1.96 \times 0.010 = 0.0196$$

The margin of error measures the distance from $\hat{p}$ to each endpoint.

If this number changes...	*Then the interval width...*
Standard error gets larger	Gets larger.
Confidence level rises from 95% to 99%	Gets larger because z* rises.

Practice 3: calculate the endpoints

Question 7 0 pts

Using the margin of error from Example 5.8, what is the 95% confidence interval?

- ◯ Just show me the answer.
- ◯ (0.877, 0.897)
- ◯ (0.0196, 1.96)
- ◯ (0.887, 0.897)
- ◯ (0.8674, 0.9066)

Practice 3: calculate the endpoints

We have $\hat{p} = 0.887$ and margin of error = 0.0196.

Step	Calculation	Result
Lower endpoint	0.887 − 0.0196	0.8674
Upper endpoint	0.887 + 0.0196	0.9066
95% confidence interval	(lower endpoint, upper endpoint)	(0.8674, 0.9066)

Practice 3: calculate the endpoints

Question 8 0 pts

Which interpretation is appropriate for Example 5.8?

- ○ 95% of individual adults are inside the interval.
- ○ Just show me the answer.
- ○ We are 95% confident that the true proportion of U.S. adults who support the policy is between about 86.7% and 90.7%.
- ○ There is a 95% probability that each individual adult supports the policy.
- ○ There is a 95% probability that this fixed parameter moves into the interval later.

Practice 4: 95% versus 99%

Question 5 0 pts

With the same standard error, what happens when we change from a 95% confidence interval to a 99% confidence interval?

- ◯ Just show me the answer.
- ◯ The standard error becomes zero.
- ◯ The critical value decreases, so the interval is narrower.
- ◯ The interval is unchanged.
- ◯ The critical value and margin of error increase, so the interval is wider.

95% versus 99%

95% confidence	z* = 1.96	Baseline interval width
99% confidence	z* = 2.58	Wider interval
Reason	Higher confidence needs a wider interval	Lower precision

Practice 6: precision and validity

A survey reports that 49% of U.S. adults find healthcare affordability difficult, with a published overall sampling margin of error of ±2.1 percentage points.

Published margin	49% ± 2.1% gives 46.9% to 51.1%. This describes precision.
Subgroups	Smaller samples usually have larger standard errors, so their estimates are less precise.
Validity	Margin of error does not solve biased wording, coverage, or an unrepresentative sample.

A narrow interval can still mislead when the measurement or sample does not represent the intended population.

Hypotheses first

A college considers a new tutoring program. The current pass rate is 70%.
Under the new program, 75 out of 100 students passed the exam.
The question: does the program change the pass rate?

Null hypothesis H_0	The program does not change the pass rate: $p = 0.70$.
Alternative hypothesis H_a	The program changes the pass rate: $p \neq 0.70$.
Evidence question	If H_0 were true, would our sample result be unusually far from 0.70?

The null gives us a chance model.
We ask whether the data look surprising under that model.

Decision errors

Type I error

Reject H_0 when the program truly has no effect. We claim a change when there is none.

Type II error

Fail to reject H_0 when the program truly changes pass rates. We miss a real change.

Correct decision

The decision matches reality. Which error would cause more harm depends on the setting.